LHCb Geometry Acceptance Studies

FT_withHole, MP, three UP variants and TV emulation

19 October 2026

Six detector setups, each evaluated for MagUp and MagDown

Main comparison: six geometries, two polarities

UP variant	FT T1/T2/T3	FT acceptance
GDML nominal	8/8/10 or 10/10/12	FT_withHole
Y8: top/bottom cutoff	8/8/10 or 10/10/12	FT_withHole
Y8X6: additional left/right cutoff	8/8/10 or 10/10/12	FT_withHole

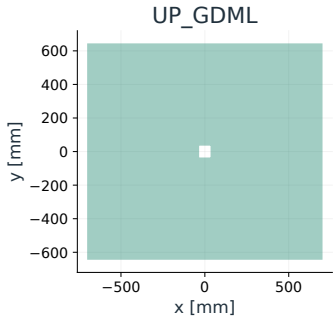
- Compare track-by-track `nFT`, `nMP`, and their combined `hasMT` cut.
- Compare `nUP` across the three UP border variants.
- Store TV cuts separately: `hasTV_3Hits`, `hasTV_4Hits`.
- Every column carries a geometry suffix and, when both are run, a polarity suffix.

Physical surfaces and counting layers

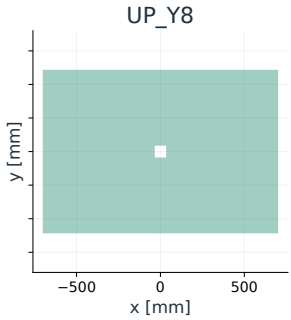
Detector	Surfaces	Logical layers	Global z [mm]
TV	100	50	-288.0–558.1
RICH1	2	2	795.0–2237.2
UP	16	4	2314.0–2656.0
MP	5	5	7665.5–9132.5
FT	12	12	7902.7–9384.7
RICH2	2	2	9487.5–11839.2
PicoCal	2	2	12565.9–12601.0
MUON	4	4	15270.0–18870.0

A logical layer fires if any of its surfaces contains the track at that surface's own z. Upstream surfaces do not fire. Multiple sensors in one module do not create multiple counted layers.

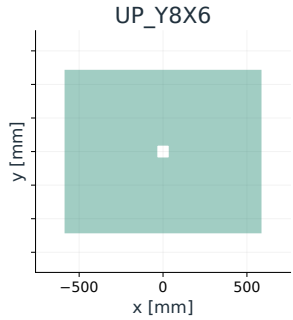
UP variants: same 16 surfaces and four counting layers



Outer: 1393.7 x 1279.8 mm
Central hole envelope: 77.3 x 79.9 mm



Outer: 1393.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm

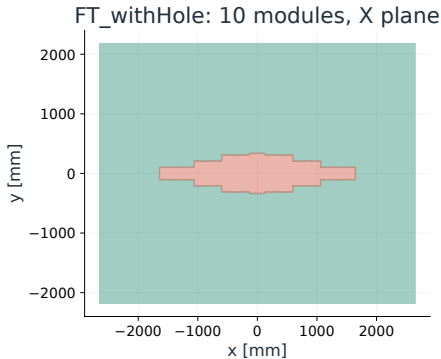


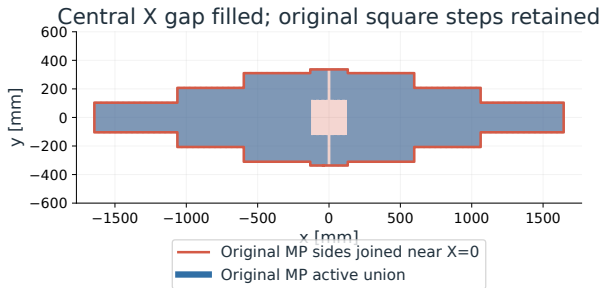
Outer: 1164.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm

Actual stored XY masks, not ideal box replacements. Crops preserve the sensor pattern and beam exclusion; dimensions and central holes are quoted above.

The four logical layer centres are 2317.5, 2372.5, 2597.5 and 2652.5 mm. Each contains four nearby source surfaces.

FT_withHole subtracts the joined MP acceptance

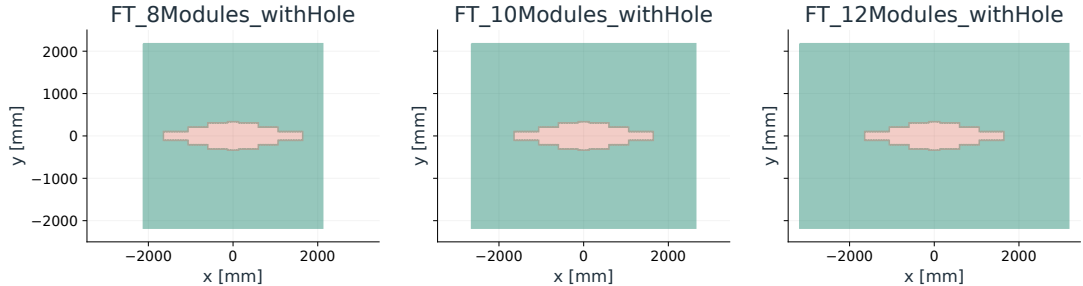




Join the MP left/right sides only across the central x gap, preserving the square outer steps. Subtract that solid envelope from every FT XY plane.

The hole spans 3288.4 $\times$ 672.6 mm. It is not a convex interpolation, and it is not projected in z.

FT widths: 8, 10 and 12 modules



Outer: 4234.4 x 4354.0 mm
Hole envelope: 3288.4 x 672.6 mm
Kept: 17.056 m2; removed: 7.15%

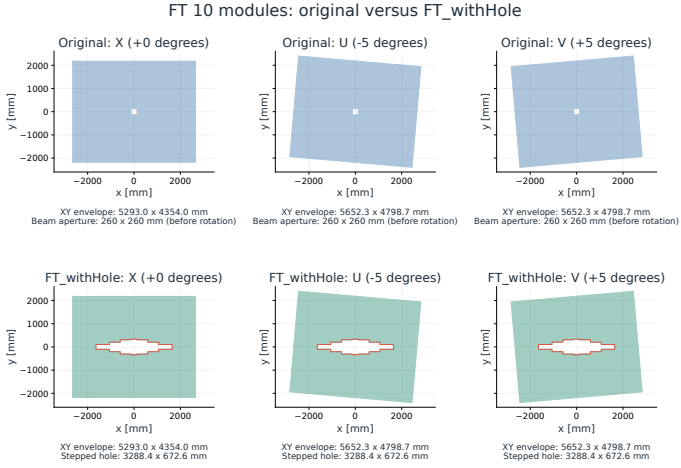
Outer: 5293.0 x 4354.0 mm
Hole envelope: 3288.4 x 672.6 mm
Kept: 21.665 m2; removed: 5.71%

Outer: 6351.6 x 4354.0 mm
Hole envelope: 3288.4 x 672.6 mm
Kept: 26.274 m2; removed: 4.76%

One module is 529.3 mm wide. Unrotated widths are 4234.4, 5293.0 and 6351.6 mm; height is 4354 mm. All three shown acceptances use the MP-derived hole.

T1/T2 compare 8 against 10 modules; T3 compares 10 against 12 modules. Both setups still have **12 FT counting layers**.

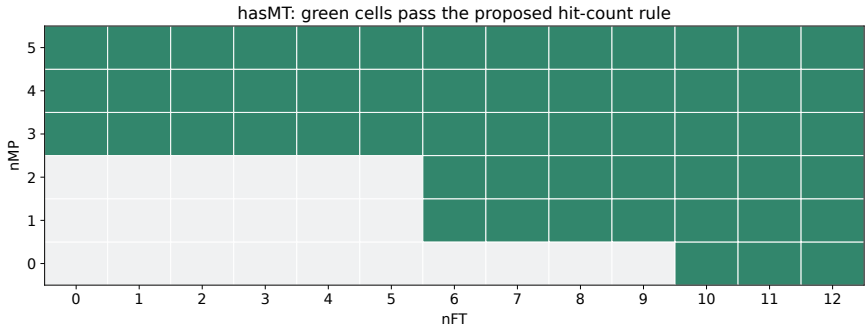
FT stereo layers retain the rotated module borders



Each station is X/U/V/X: $0, -5, +5, 0$ degrees around z. The rectangular module crop follows the stereo rotation. The MP hole is then subtracted in the global XY frame.

Configurable MT requirement

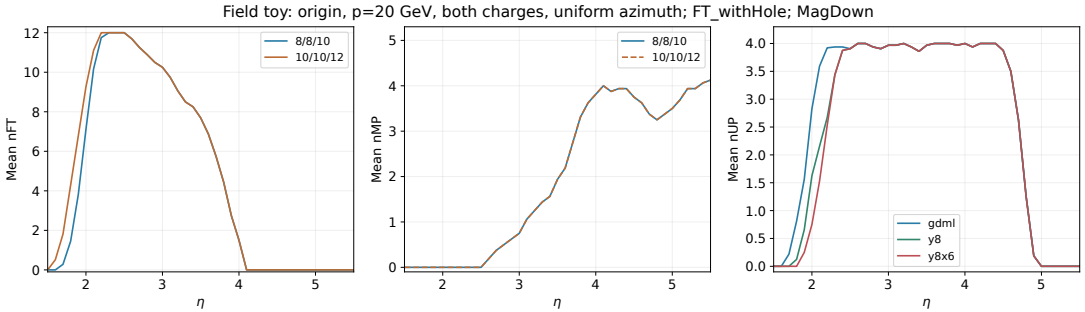
$$\texttt{hasMT} = (n_{\mathrm{MP}} \geq 3) \ \vee \ (n_{\mathrm{FT}} \geq 10) \ \vee \ \big[(n_{\mathrm{MP}} \geq 1) \wedge (n_{\mathrm{FT}} \geq 6)\big]$$



Thresholds are configurable globally or per geometry with `MTSelection`. Keep the counts to change cuts offline without transport.

TV and UP requirements remain separate from `hasMT`.

Field-propagated hit-count toy

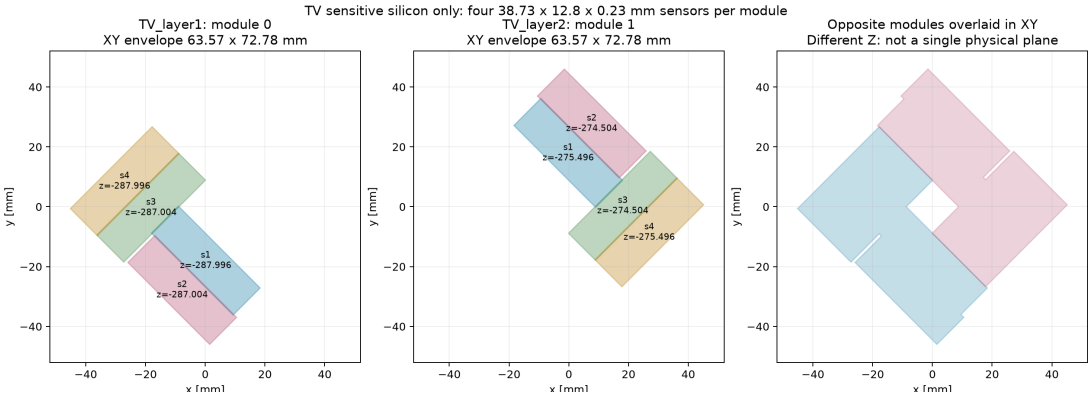


2,624 origin tracks, $p = 20$ GeV, equal charges, 41 η points and 32 azimuth directions; no material. Both polarities were evaluated.

Checked: n_{MP} is unchanged by the FT width choice; baseline $n_{\mathrm{FT}} \geq$ lower n_{FT}; UP crops never add hits.

Illustrative geometry checks only: these are not efficiencies for an analysis sample.

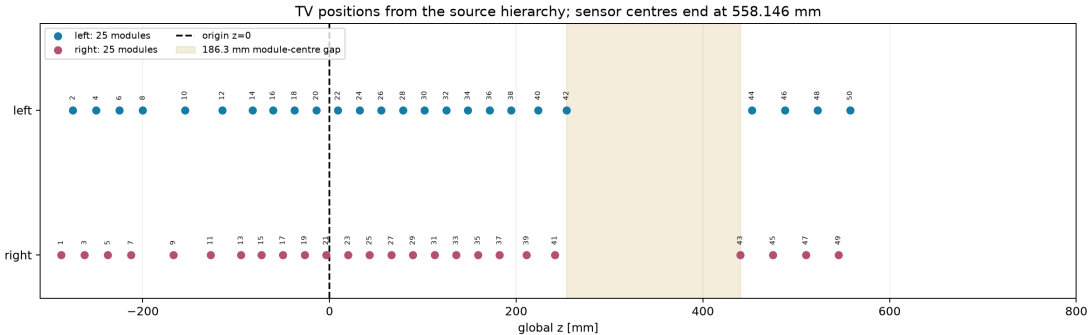
TV acceptance comes from sensitive silicon



Select `TV:SensorSilicon` / `lvDet` in the source ROOT hierarchy: 200 sensors, 38.73 × 12.8 × 0.23 mm each.

Four sensors form one module, at two centre-z positions 0.992 mm apart. Only coplanar pairs are unioned; actual sensor gaps remain inactive.

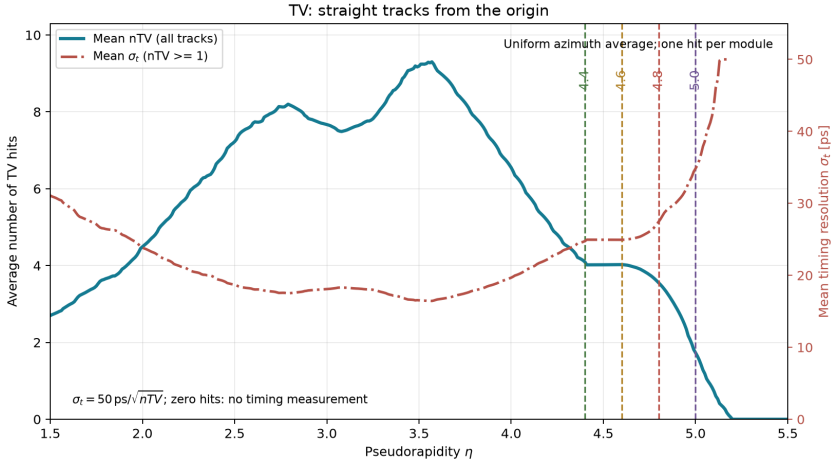
TV positioning: 100 sensor planes, 50 modules



Sensor centres span −287.996 to 558.146 mm. There is no sensitive silicon around 700 or 780 mm in this input geometry.

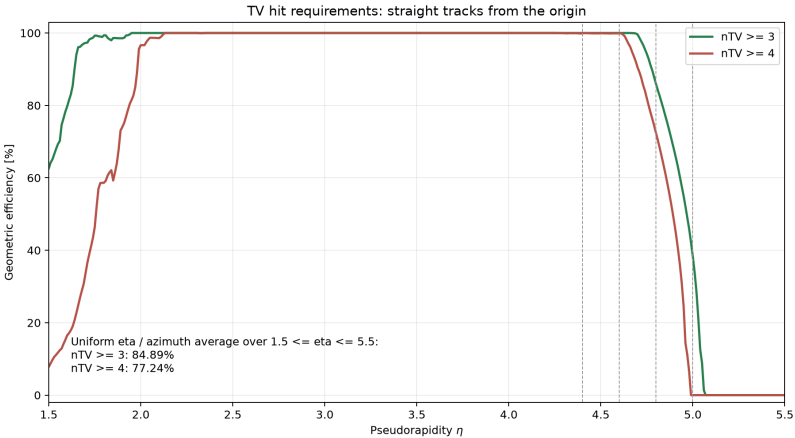
From the origin: 21 upstream modules cannot fire; 29 are reachable. The 186.3 mm downstream gap is present in the ROOT hierarchy.

TV emulation: mean hits and expected timing



Straight origin rays, uniform azimuth. Timing uses $\sigma_t = 50\,\mathrm{ps}/\sqrt{n_{\mathrm{TV}}}$ per track, averaged for $n_{\mathrm{TV}} \geq 1$. Zero-hit tracks have no timing measurement.

TV geometric efficiency for three and four hits



$1.5 \leq \eta \leq 5.5$

Uniform η and azimuth averages:

$n_{\mathrm{TV}} \geq 3$: **84.89%**

$n_{\mathrm{TV}} \geq 4$: **77.24%**

1,154,880 rays.

Numerical azimuth refinement changes efficiencies by less than 0.16 percentage points.

No field, vertex spread, sensor inefficiency or reconstruction in this TV toy. Physics-sample efficiencies need the actual track distribution.

Field and transport validation

- States are $(x, y, t_x, t_y, q/p)$ at a given z; positions in mm, momentum in MeV/c, field in tesla.
- Read Bx/By/Bz TH3 histograms; integrate with vectorized RK4. No material scattering, energy loss or detector response.
- Bundled reference map: $\int_0^{10000\,\mathrm{mm}} B_y\,dz = -4.1856$ T m. It is designated MagDown; MagUp reverses *all three* components.
- One transport per polarity is reused across six geometries: two transports, not twelve. Counts and cuts keep geometry and polarity suffixes.
- Tests cover straight rays, charge/field reversal, upstream exclusions, counting groups, FT/UP border changes, MT cut boundaries and failed transport.
- **118 tests passed** with the local ROOT field map, including the two-polarity equivalence and track-order checks.

Sign-reversed maps are an idealization, not two independently measured maps.

One call labels every geometry and both polarities

```
from xyshapes_detectors import PolarityComparison, MTSelection

service = PolarityComparison.from_files(
    "MagField/magnetic_field_v5r0.root",
    map_polarity="magdown",
    mt_selection=MTSelection(3, 10, 1, 6),
)
flags = service.label_tracks(states)
```

Example output columns:

```
nFT_up_y8x6_ft_8_8_10_withHole_magup
nMP_up_y8x6_ft_8_8_10_withHole_magup
hasMT_up_y8x6_ft_8_8_10_withHole_magup
```

Equivalent MagDown columns are retained. The flags do not determine which polarity was used in the data; choose that using the ntuple's metadata.

Using the masks in decay studies

- The field map, reusable masks, loader, transport, cuts and tests are in the repository. No GDML extraction or ROOT installation is needed to consume them.
- Reuse one service across bounded chunks. Prefix flags by daughter name; preserve candidate rows and jagged track offsets when writing new branches.
- Compare joint $(n_{\mathrm{FT}}, n_{\mathrm{MP}})$ distributions and `hasMT` between 8/8/10 and 10/10/12, holding UP and polarity fixed.
- Compare n_{UP} for GDML, Y8 and Y8X6, holding FT and polarity fixed. Combine with the chosen TV cut only at analysis level.
- Next physics step: run on the selected decay ntuples, retain their weights and denominator, and evaluate daughter and decay-level efficiencies.

Reproduction: `examples/create_acceptance_study_slides.py`.
Usage: `docs/decay_study_quickstart.md`; API: `docs/geometry_comparison.md`.